Experiment 12

Title

Unsupervised Pattern Recognition using K-Means Clustering on 2D Data

SAP ID:	590013857
Dataset:	https://www.kaggle.com/datasets/neurocipher/kmeans-clustering-2d-dataset/data

AIM

To apply K-Means clustering on a synthetic 2D dataset, analyze the effect of different values of K, and determine the optimal number of clusters using evaluation metrics.

THEORY

K-Means is an unsupervised learning algorithm used to partition data into K clusters. It works by minimizing the Within-Cluster Sum of Squares (WCSS).

Key steps:

- Initialize K centroids
- Assign data points to nearest centroid
- Update centroids
- Repeat until convergence

Cluster evaluation metrics:

- **WCSS**: Measures compactness (lower is better)
- **Silhouette Score**: Measures separation (-1 to 1, higher is better)
- **Davies-Bouldin Index**: Lower is better
- **Calinski-Harabasz Index**: Higher is better

Commands/Code Used

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

kmeans = KMeans(n_clusters=3)
labels = kmeans.fit_predict(X)

score = silhouette_score(X, labels)
```

Results

Synthetic Dataset Visualization

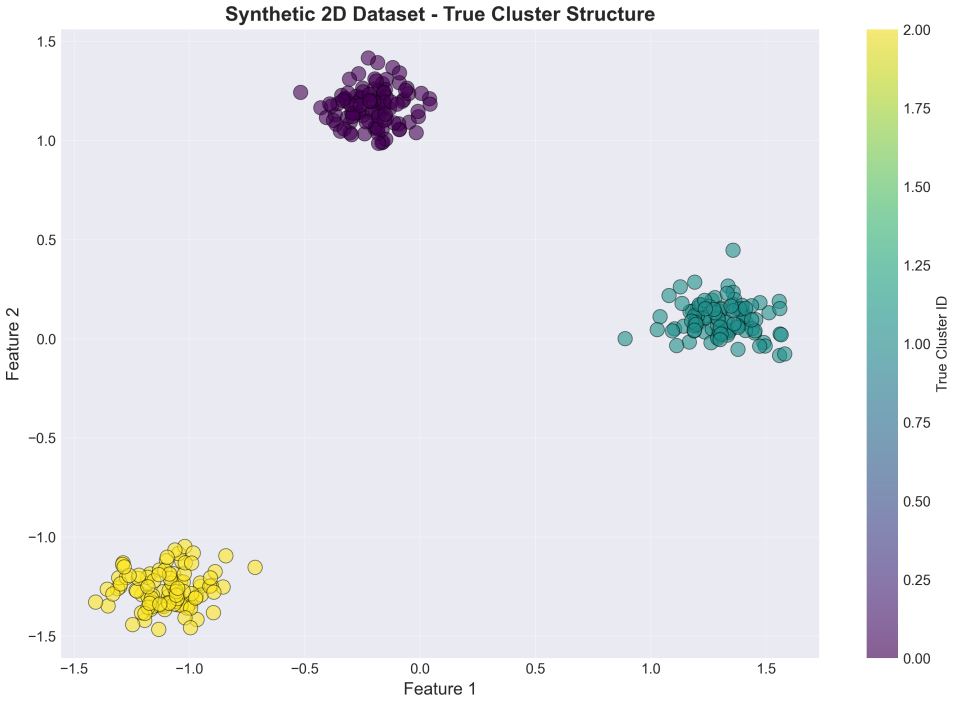

Figure 1: True cluster structure in dataset

Elbow Method and Metrics

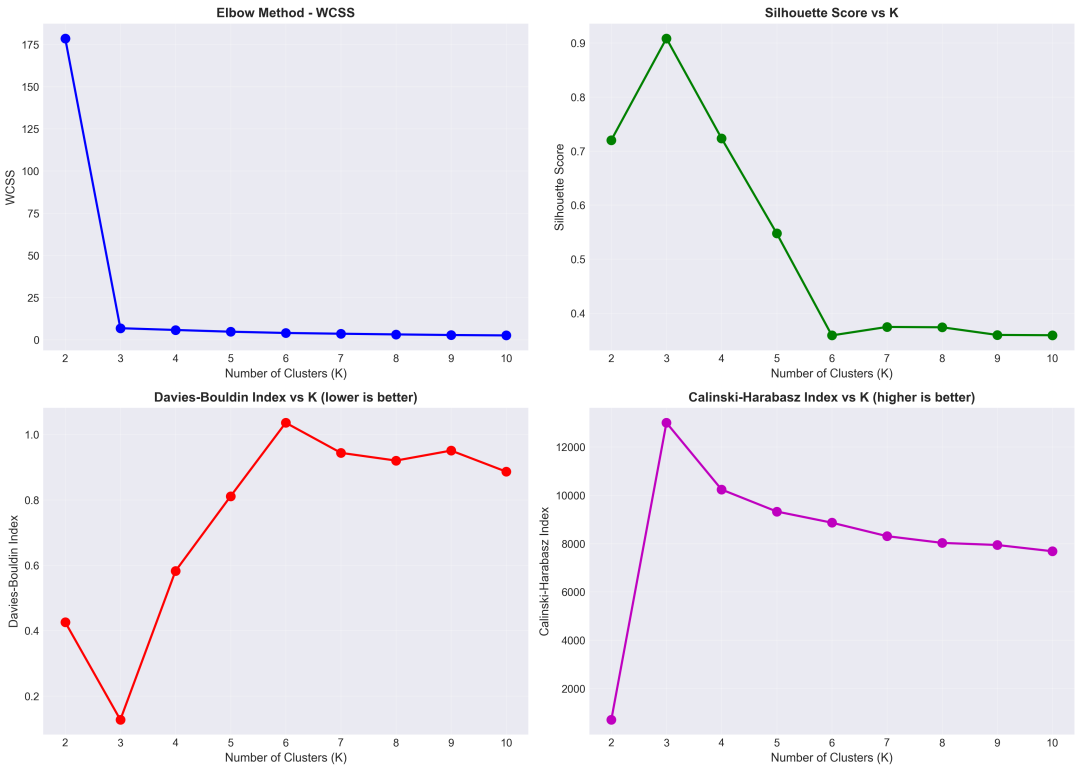

Figure 2: WCSS, Silhouette, Davies-Bouldin, and Calinski-Harabasz analysis

Effect of Different K Values

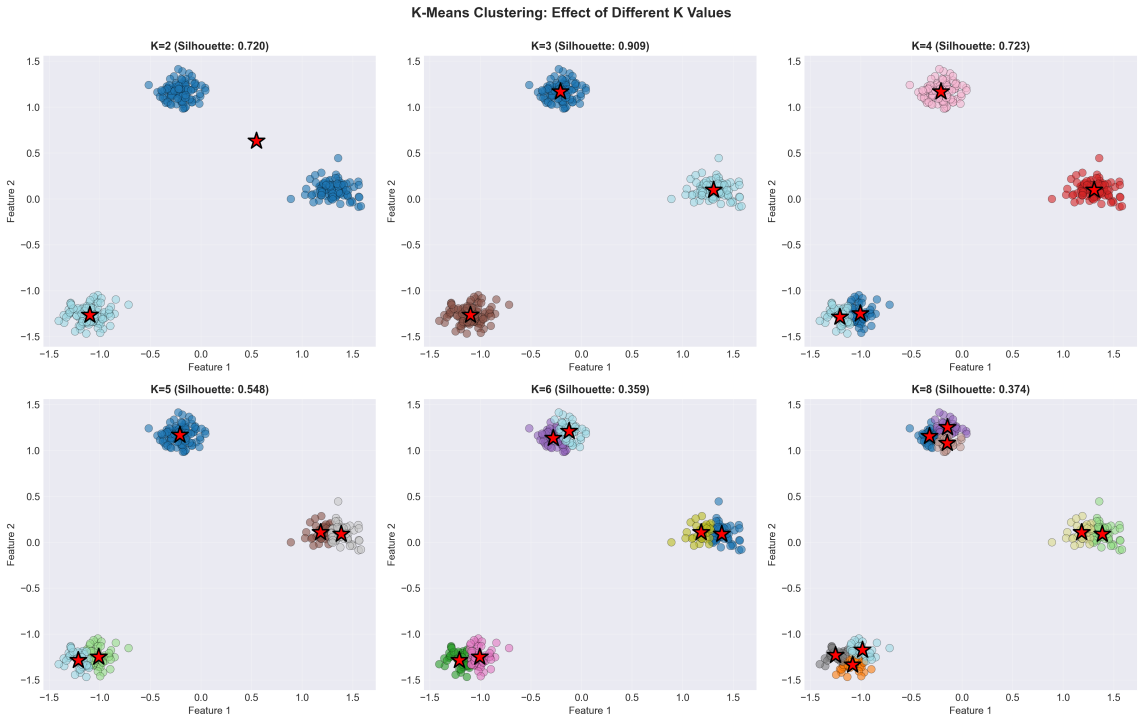

Figure 3: Clustering results for different K values

Final Clustering (K=3)

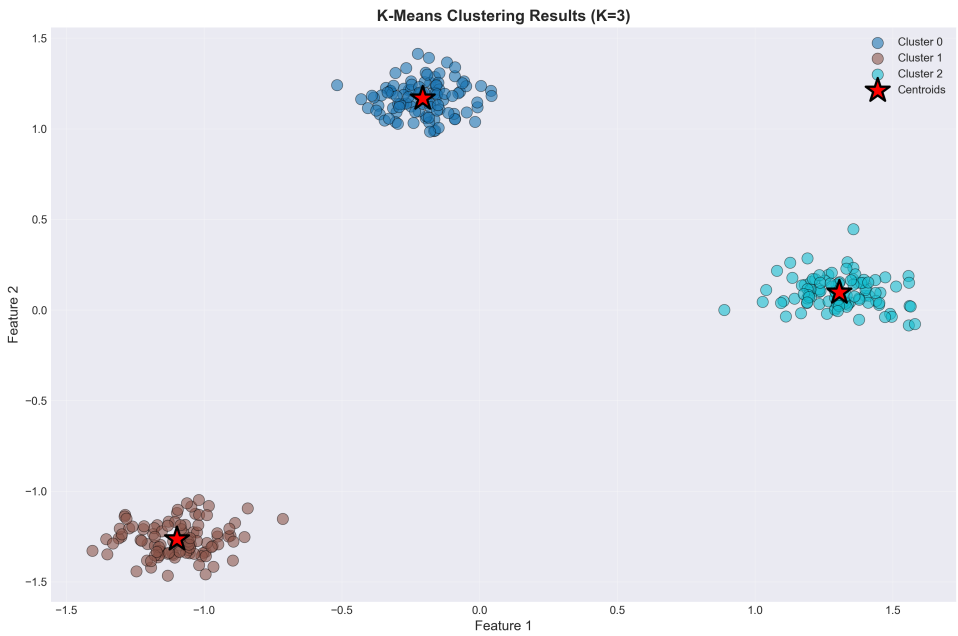

Figure 4: Final K-Means clustering with centroids

Analysis

- Elbow method shows sharp drop at K=3
- Silhouette score is highest at K=3 (0.91)
- Davies-Bouldin index is minimum at K=3
- Calinski-Harabasz index is maximum at K=3

Thus, all metrics agree that:

$$K = 3 is the optimal number of clusters$$

Conclusion

The K-Means algorithm successfully identified the natural cluster structure in the dataset. The optimal number of clusters was found to be 3 using multiple evaluation metrics.

The experiment demonstrates that:

- K-Means is effective for well-separated data
- Proper selection of K is critical
- Multiple evaluation metrics provide reliable validation